

Task List for Airbnb Data Analysis

TrajectAI ML and AI Bootcamp, Week 2: Python's Data Toolkit

- Q.1)** Import the essential libraries for data analysis.
- Q.2)** Load the Airbnb dataset from a CSV file into a pandas DataFrame.
- Q.3)** Print the first 20 rows of the dataset to get an initial understanding of its structure and content.
- Q.4)** Find out the shape of the dataset and explain what it indicates.
- Q.5)** List all column names and replace spaces in column labels with underscores for consistency and ease of use.
- Q.6)** Check the data types of each attribute to understand what type of data each column holds.
- Q.7)** Convert the `minimum_nights` and `price` columns from float to integer after removing any non-numeric characters such as currency symbols. Convert `last_review` to a datetime type.
- Q.8)** Share the information of the dataset. What is its importance in EDA?
- Q.9)** Provide summary statistics for numerical columns. Highlight any unusual calculation you find.
- Q.10)** Provide summary statistics for categorical columns. Highlight any unusual calculation you find.
- Q.11)** Identify duplicate rows and remove them to ensure data integrity. Check exact duplicates, business key duplicates on `id`, and near duplicates.
- Q.12)** Check each column for null or empty values, and test whether the missingness is patterned rather than random.
- Q.13)** Replace null values with suitable replacements, choosing the fill from the cause of the gap rather than from a global statistic. Add a missing indicator for every column you impute.
- Q.14)** Drop the columns `last_review`, `name`, and `id`, after first extracting the features they contain.
- Q.15)** Find the number of unique values for `room_type`, `neighbourhood_group`, and `neighbourhood`. Highlight the unusual answer and fix it. Write the fix so that it would also catch a typo you have not seen yet.
- Q.16)** What are the top 10 host IDs with the highest number of bookings? State which column you are using as a proxy for bookings, and why.
- Q.17)** Identify the top 10 host names by number of bookings and create a bar chart. State why a host name is not a safe grouping key.
- Q.18)** What room types does the host with the highest number of bookings offer, and what is the price range?
- Q.19)** How is listing supply split across room types, and how does that split differ by borough?

- Q.20)** Identify the neighbourhood group with the highest number of bookings and visualize it with a pie chart. Say when a pie chart is the wrong choice.
- Q.21)** Calculate the average price of all listings. Report a second statistic that describes the typical listing better, and justify it.
- Q.22)** Determine the minimum and maximum prices of the listings, and inspect the rows they come from.
- Q.23)** Compare the average prices of different room types to see how they vary. Include a measure of spread, not only the means.
- Q.24)** Identify the neighbourhood group with the highest number of reviews and highlight any interesting findings. Answer for total reviews and for reviews per listing, and explain why the two rankings differ.
- Q.25)** Move a column into the index with `set_index` and back out with `reset_index`. State when each form is the one you need.
- Q.26)** Build a borough by room type pivot table of median price. Add margins, then explain why the "All" row is not the average of the rows above it.
- Q.27)** Cross-tabulate borough against room type three ways, using `normalize="index"`, `"columns"`, and `"all"`. Write the sentence each table answers.
- Q.28)** Build a wide summary table of three metrics per borough, melt it to long form, chart it, then pivot it back. Explain the difference between `pivot` and `pivot_table`.
- Q.29)** Build a small borough lookup table and merge it onto the listings. Compare `how="left"` against `how="inner"`, check the row count before and after, and use `validate=` to make pandas catch a non-unique key for you.
- Q.30)** Stack two subsets with `concat` on both axes. Explain when `concat` is the right tool and when you actually wanted `merge`.
- Q.31)** Use `groupby().transform()` to add each listing's borough median price as a column, then compute the listing's price as a ratio of it. Explain why `agg` cannot do this, and why the resulting feature is dangerous.
- Q.32)** Resample `last_review` by month and plot listing counts and median price over time. Identify at least one artifact of the snapshot in the resulting series.
- Q.33)** Write a business rule table giving the legal range of every numeric column, then report every violation. Handle `price = 0`, `minimum_nights` above 365, and coordinates outside the New York bounding box. Justify a different response for each.
- Q.34)** Detect outliers with the IQR rule, the z score rule, and percentile capping. Compare what each method flags and decide, per column, whether to drop, clip, flag, or transform.
- Q.35)** Write cross-field consistency checks: reviews greater than zero must have a review date, `reviews_per_month` must be zero when the review count is zero, and the provided host listing count should match the count you observe. For every check that fails, decide whether it is a broken record or a definition mismatch.
- Q.36)** Optimize memory: convert low cardinality strings to the `category` dtype and downcast numerics. Report the before and after footprint, and name one operation that gets harder once a column is categorical.

- Q.37)** Split your cleaning into `fit_cleaning_params(train)` and `apply_cleaning(df, params)`, then prove that the test set is cleaned using only training statistics. Finish with a `validate_clean_data` gate that returns a list of problems and is empty when the table is fit to use.
- Q.38)** Build a borough by room type pivot table for both median price and listing count, and visualize both as heatmaps. Quantify whether the Manhattan premium is the same for every room type.
- Q.39)** Produce a Spearman correlation matrix against log price. Rank the features by their correlation with the target, and list every feature pair correlated above 0.6. Explain why Spearman rather than Pearson on this data.
- Q.40)** Map the listings. Plot them coloured by borough, coloured by log price, and as a hexbin of mean price. Then engineer haversine distance features to Times Square, Wall Street, Central Park, and JFK, and show that they correlate with price far better than raw latitude and longitude.
- Q.41)** Analyse the `availability_365` distribution. Explain the spikes at 0 and 365 and compare the price, reviews, and minimum nights of each segment.
- Q.42)** Band `minimum_nights` into 1 night, 2 to 3, 4 to 7, 8 to 29, and 30 or more, and compare price, reviews, availability, and the entire home share across bands.
- Q.43)** Segment hosts by portfolio size. Show what share of listings the professional hosts control, then compare a random train test split against a `GroupShuffleSplit` on `host_id` and count how many test rows share a host with the training set.
- Q.44)** Extract features from the listing name: length, word count, capitalisation ratio, keyword flags for luxury, budget, location, and amenity terms, and the bedroom count parsed from the title. Measure the median price lift of each keyword group.
- Q.45)** Write ONE `engineer_features` function producing distance, recency, availability, host, stay type, engagement, and neighbourhood aggregate features. Rank the results by correlation with the target.
- Q.46)** Encode the categorical columns three ways: one hot for low cardinality, frequency encoding, and smoothed target encoding fitted on training data only. Show that smoothing pulls rare categories towards the global mean.
- Q.47)** Write a leakage detector and use it to identify features derived from the target. Explain why `neighbourhood_median_price` is acceptable while `host_median_price` and `price_vs_neighbourhood` are not.
- Q.48)** Test whether the `has_reviews` missing indicator carries real signal, using a group comparison and a rank based effect size. Keep it only if it earns its place.